

Bálint Gyevnár

Carnegie Mellon University, Pittsburgh, PA | bgyevnar@cmu.edu | gbalint.me

RESEARCH INTERESTS

Metascience; AI Scientists; Agentic Scientific Discovery; Explainable Multi-Agent Systems; Responsible AI; AI Safety; Human-AI Interaction.

EDUCATION

PhD in Natural Language Processing Sep. 2021 – Jan. 2026

University of Edinburgh

Edinburgh, UK

Thesis: Action Explanation of Multi-Agent Systems via Counterfactual Reasoning

Supervisors: Stefano V. Albrecht, Shay B. Cohen, and Christopher G. Lucas

Integrated Master of Informatics Sep. 2016 – May 2021

University of Edinburgh

Edinburgh, UK

Thesis: Comparison of Account Survival on Twitter During the First Wave of COVID-19

Supervisor: Maria Wolters

Exchange Year in Computer Science Aug. 2018 – Jun. 2019

Nanyang Technological University

Singapore

ACADEMIC EXPERIENCE

Postdoctoral Research Associate Sep. 2025 – present

Carnegie Mellon University

Pittsburgh, PA

- Investigating **failure modes of agentic AI scientists**, such as data poisoning, reward hacking, spin doctoring, with actionable mitigations—work with *Nihar Shah* and *Atoosa Kasirzadeh*;
- Understanding the epistemic **landscape of AI hypothesis generation** and steering, e.g. using formalization of mathematics in Lean—work with *Simon DeDeo* and *David Barack*.

Teaching Assistant Sep. 2020 – May 2025

University of Edinburgh

Edinburgh, UK

- Teaching assistant for courses in machine learning, computing systems, and two data analysis courses; Marker for natural language processing, reinforcement learning, machine learning courses.

Research Internship May 2020 – Oct. 2020

Five AI Ltd.

Edinburgh, UK

- Development and evaluation of goal-based interpretable prediction and planning for autonomous vehicles;
- Main contributor of open-source implementation on GitHub with added support for CARLA.

SERVICE

Academic Reviewing

- *Program committee member:* AAAI 2025, AIES 2025, FAccT 2026, XAI 2025;
- *Reviewer:* Nature Communications 2025, Artificial Intelligence Review 2025, International Journal of Human-Computer Interaction 2025, Transactions on Intelligent Transportation Systems 2025, CHI 2026, ECAI 2023, EMNLP2025, ICRA 2025, IASEAI 2026, IROS 2025, NeurIPS 2025.

TEACHING

- *Teaching Assistant (Spring 2024)*: Evaluating Sustainable Lands and Cities; Data, Mobility, Infrastructure;
- *Marker (Spring 2022-24)*: Machine Learning Theory, Reinforcement Learning, Natural Language Processing
- *Tutor (Autumn 2020)*: Introductory Applied Machine Learning; Introduction to Computing Systems;

AWARDS

- Colours Award for Outstanding Volunteering Contribution to Sports;
Edinburgh University Sports Union; Jun. 2024.
- Love, Sex, and AI — AI100 Early Career Essay Competition Featured Essay;
One Hundred Year Study on Artificial Intelligence at Stanford University; Aug. 2023.
- £4,000 GBP Trustworthy Autonomous Systems Early Career Researcher Award;
UK Research & Innovation; Jun. 2023.
- \$1,000 USD Shape the Future of ITS Competition Award;
IEEE Intelligent Transportation Systems Society; Aug. 2022.

PUBLICATIONS

Poisoned Public Datasets Steer Autonomous AI Research in Arbitrary Directions
Forthcoming; email for draft.

B. Gyevnár, A. Kasirzadeh, N.B. Shah

Bridging the Gap in the Responsible AI Divides

arXiv preprint 2603.14495, 2026;

B. Gyevnár, A. Kasirzadeh.

Integrating Counterfactual Simulations with Language Models for Explaining Multi-Agent Behaviour

25th International Conference on Autonomous Agents and Multiagent Systems, AAMAS 2026;

B. Gyevnár, C.G. Lucas, S.V. Albrecht, S.B. Cohen.

AI Safety for Everyone

Nature Machine Intelligence, 2025;

B. Gyevnár, A. Kasirzadeh.

Objective Metrics for Human-Subjects Evaluation in Explainable Reinforcement Learning

Multi-Disciplinary Conference on Reinforcement Learning and Decision Making, RLDM 2025;

B. Gyevnár, M. Towers.

People Attribute Purpose to Autonomous Vehicles When Explaining Their Behavior: Insights From Cognitive Science for Explainable AI

ACM Conference on Human Factors in Computing Systems, CHI 2025;

B. Gyevnár, S. Droop, T. Quillien, S.B. Cohen, N.R. Bramley, C.G. Lucas, S.V. Albrecht.

Causal Explanations for Sequential Decision-Making in Multi-Agent Systems

23rd International Conference on Autonomous Agents and Multiagent Systems, AAMAS 2024;

B. Gyevnár, C. Wang, S.B. Cohen, C.G. Lucas, S.V. Albrecht.

Explainable AI for Safe and Trustworthy Autonomous Driving: A Systematic Review

IEEE Transactions on Intelligent Transportation Systems, 2024;

A. Kuznietsov*, B. Gyevnár*, C. Wang, S. Peters, S.V. Albrecht. [* equal contribution]

Bridging the Transparency Gap: What Can Explainable AI Learn From the AI Act?

26th European Conference on Artificial Intelligence, ECAI 2023;

B. Gyevnár, N. Ferguson, B. Schafer.

Deep Reinforcement Learning for Multi-Agent Interaction

AI Communications, 35(4), 357-368, 2022;

I.H. Ahmed, C. Brewitt, I. Carlucho, F. Christianos, M. Dunion, E. Fosong, S. Garcin, S. Guo, B. Gyevnár, T. McInroe, G. Papoudakis, A. Rahman, L. Schäfer, M. Tamborski, G. Vecchio, C. Wang, S.V. Albrecht.

Communicative Efficiency or Iconic Learning: Do Communicative and Acquisition Pressures Interact to Shape Colour-Naming Systems?

Entropy, 24(11), 1542, 2022;

B. Gyevnár, G. Dagan, C. Haley, S. Guo, F. Mollica.

A Human-Centric Method for Generating Causal Explanations in Natural Language for Autonomous Vehicle Motion Planning [best paper runner-up]

Workshop on Artificial Intelligence for Autonomous Driving, IJCAI 2022;

B. Gyevnár, M. Tamborski, C. Wang, C.G. Lucas, S.B. Cohen, S.V. Albrecht.

Interpretable Goal-based Prediction and Planning for Autonomous Driving

International Conference on Robotics and Automation, ICRA 2021;

S.V. Albrecht, C. Brewitt, J. Wilhelm, B. Gyevnár, F. Eiras, M. Dobre, S. Ramamoorthy.

GRIT: Fast, Interpretable, and Verifiable Goal Recognition with Learned Decision Trees for Autonomous Driving

IEEE/RSJ International Conference on Intelligent Robots and Systems, IROS, 2021;

C. Brewitt, B. Gyevnár, S. Garcin., S.V. Albrecht.

INVITED TALKS

Human and AI solution paths in formalizing expert mathematics

Social Reasoning and the Ecology of Thought, IVADO & MILA, 2026.

AI safety for everyone

Talk at the 9th Workshop of the Center for Human-Compatible AI, 2025.

How do we make explainable AI work for people?

Talk at the Machine Learning and Modelling Seminar, Charles University of Prague, 2024.